SSJ approach and toolkit

Matthew Rognlie

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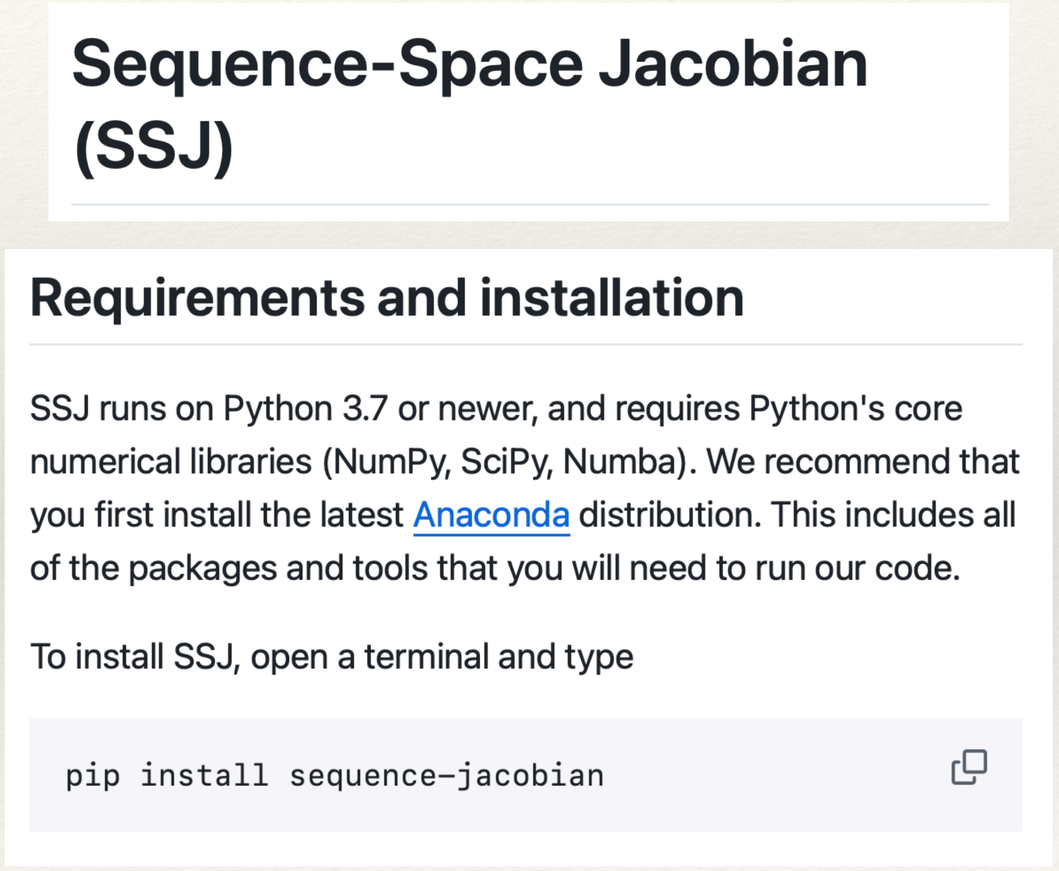
This lecture

- So far: built SIM model and HANK, applied to fiscal and monetary policy
- Until now, we've written our own code for every step:
  - Solved for steady state
  - Implemented fake news algorithm to obtain Jacobians
  - Hand-derived Jacobian system, solved to obtain impulse responses
- Today, introduce the **SSJ toolkit** to automate much of this process

What is the SSJ toolkit?

- Started in 2019 as code accompanying the Sequence-Space Jacobian paper
- Major revisions in 2020, 2021, 2022 to turn it into a bona fide "toolkit"
- Less active development in recent years, but planning for big push soon
  - Lots of new and improved methods to incorporate!



(see github.com/shade-econ/sequence-jacobian)

Why is a toolkit useful?

- Many computations are **common across models**:
  - Het-agent models: lotteries for distribution, backward and forward iteration for steady state, fake news algorithm for Jacobians, …
  - All models: differentiate equations, write as a linear system in sequence space, apply chain rule, solve the system, …
- Coding these manually gets tedious as models become more complex, and **errors become more likely!** [Even now with AI…]
- Toolkit philosophy: automate everything **that's easy to automate**, leave some **high-level work to the user**
- See also related toolkits: HARK, BASEforHANK, econpizza, and more

SSJ toolkit essentials

Concept 1: blocks

- Toolkit is built around **blocks**
  - Each block is a piece of an economic model
  - It's a mapping from aggregate **inputs** to aggregate **outputs**
- Four main kinds of blocks:
  - **SimpleBlock**: simple aggregate equations, can have time leads or lags
  - **HetBlock**: heterogeneous agents that collectively take in some aggregate inputs (e.g. interest rates) and produce some outputs (e.g. consumption)
  - **SolvedBlock**: wraps around another block, solves for "unknowns" given "targets"
  - **CombinedBlock**: composition of many blocks

Example: SimpleBlock

- A SimpleBlock is defined like an ordinary Python function, decorated with @sj.simple
- Can put Dynare-style time offsets in parentheses: here, production is $Y_t = A_t K_{t-1}^{\alpha} L_t^{1-\alpha}$
- Block inputs are the function inputs; outputs must be listed in the return statement

```
import sequence_jacobian as sj
@sj.simple
def prod(A, K, L, alpha):
    Y = A * K(-1)**alpha * L**(1 - alpha)
    MPK = alpha * Y / K(-1)
    MPL = (1 - alpha) * Y / L
    return Y, MPK, MPL
```

```
print(prod), print(prod.outputs), print(prod.inputs)
```

✓ 0.0s

```
<SimpleBlock 'prod'>
['Y', 'MPK', 'MPL']
['A', 'K', 'L', 'alpha']
```

Concept 2: methods

- Each block has four standard **methods** that can be called on it:
  - **.steady_state()**
  - **.impulse_linear()**
  - **.impulse_nonlinear()**
  - **.jacobian()**
- Each also has a "solve" variant, e.g. **.solve_steady_state()** solves for steady state "unknowns" to hit certain steady-state "targets"

.steady_state() on our example

- .steady_state() is called on a dict giving the steady-state value of every input to a block, and returns a dict that adds every steady-state output
- For a SimpleBlock, it essentially evaluates the function, ignoring time offsets
- For other blocks, it's more complicated.
- Returns a SteadyStateDict, which is mostly a dict but with some enhancements (e.g. stores extra "internal" data for HetBlocks)

```
calibration = {'A': 1, 'K': 4, 'L': 1, 'alpha': 0.3}
ss = prod.steady_state(calibration)
ss
```

✓ 0.0s

```
<SteadyStateDict: ['A', 'K', 'L', 'alpha', 'Y', 'MPK', 'MPL']>
```

```
dict(**ss)
```

✓ 0.0s

```
{'A': 1,
 'K': 4,
 'L': 1,
 'alpha': 0.3,
 'Y': 1.515716566510398,
 'MPK': 0.11367874248827985,
 'MPL': 1.0610015965572785}
```

.solve_steady_state() on our example

- We might not know every steady-state parameter, but instead want to hit some calibration targets
- Then, we use .solve_steady_state(), specifying the "unknown" parameters and the "targets" we want to hit
- It's a thin wrapper around .steady_state(), with an outer loop that does a nonlinear solver
  - can also do this manually, and sometimes should to have more control!

```
ss = prod.solve_steady_state(calibration,
                             unknowns={'A':1., 'alpha': 0.5},
                             targets={'Y': 1, 'MPK': 0.1})
```

```
dict(**ss)
```

✓ 0.0s

```
{'A': np.float64(0.5743491774985149),
 'K': 4,
 'L': 1,
 'alpha': np.float64(0.40000000000002894),
 'Y': 1.0000000000000355,
 'MPK': 0.10000000000001079,
 'MPL': 0.5999999999999923}
```

.jacobian() on our example

- .jacobian() returns the sequence-space Jacobians of some outputs vs. inputs, as a **JacobianDict** (a nested dict with some enhanced features)
- [In most cases, this is numerical up to some horizon "T", but for SimpleBlocks the default is to exploit the special structure of the Jacobian, returning as a "SimpleSparse" (which can act like a regular matrix)]

```
J = prod.jacobian(ss, inputs=['K', 'L'],
                  outputs=['Y', 'MPL'])
J
```

✓ 0.0s

```
<JacobianDict outputs=['Y', 'MPL'], inputs=['K', 'L']>
```

```
J['Y', 'K']
```

✓ 0.0s

```
SimpleSparse({(-1, 0): 0.100})
```

```
J['Y', 'K'].matrix(5)
```

✓ 0.0s

```
array([[0. , 0. , 0. , 0. , 0. ],
       [0.1, 0. , 0. , 0. , 0. ],
       [0. , 0.1, 0. , 0. , 0. ],
       [0. , 0. , 0.1, 0. , 0. ],
       [0. , 0. , 0. , 0.1, 0. ]])
```

Applying JacobianDicts to impulses

- A JacobianDict can operate on impulses to its inputs (given as a dict), and calculate the resulting impulses to outputs
- The result is an ImpulseDict, again a kind of enhanced dict
- Can add and subtract ImpulseDicts, multiply them by scalars, multiply them by additional JacobianDicts, etc.

```
dL = 0.5 * 0.9**np.arange(25)
dK = 2 * 0.5**np.arange(25)
impulse = J @ {'K': dK, 'L': dL}
impulse
```

✓ 0.0s

```
<ImpulseDict: ['K', 'L', 'Y', 'MPL']>
```

```
impulse['K'][:5], impulse['Y'][:5]
```

✓ 0.0s

```
(array([2.    , 1.    , 0.5   , 0.25 , 0.125]),
 array([0.3    , 0.47   , 0.343  , 0.2687 , 0.22183]))
```

Or get in one shot with impulse methods

- **.impulse_linear()** is a more direct route to what we just did: rather than calculating the Jacobian and applying it to an impulse, it directly obtains the linearized output response

- **.impulse_nonlinear()** is the same, but does not linearize: it calculates the fully nonlinear impulse to outputs from the impulse to inputs

```
impulse2 = prod.impulse_linear(ss,
    inputs={'K': dK, 'L': dL}, outputs=['Y', 'MPL'])
impulse2['Y'][:5]
```

✓ 0.0s

```
array([0.3    , 0.47   , 0.343  , 0.2687 , 0.22183])
```

```
impulse3 = prod.impulse_nonlinear(ss,
    inputs={'K': dK, 'L': dL}, outputs=['Y', 'MPL'])
impulse3['Y'][:5]
```

✓ 0.0s

```
array([0.2754245 , 0.46979683, 0.3408179 , 0.26312152, 0.21467539])
```

[The impulse_linear response is exactly the same as on the previous slide. The impulse_nonlinear response is a bit smaller, because the shocks are huge and trigger nonlinearities.]

Concept 3: combining and solving blocks

- The toolkit becomes more powerful once we start **combining blocks**
- This composes them together, so that one block's output can be another's input
- The resulting "CombinedBlock" is treated as a single block that produces all outputs
- Here, we compose our "prod" block with a labor supply block and a block representing labor market equilibrium (setting the wage equal to MPL)

```
@sj.simple
def labor_supply(Lbar, w):
    L = Lbar * w**0.5
    return L

@sj.simple
def labor_mkt_clearing(w, MPL):
    labor_mkt = w - MPL
    return labor_mkt

model = sj.combine([prod, labor_supply, labor_mkt_clearing],
                   name='model')
model
```

✓ 0.0s

```
<CombinedBlock 'model'>
```

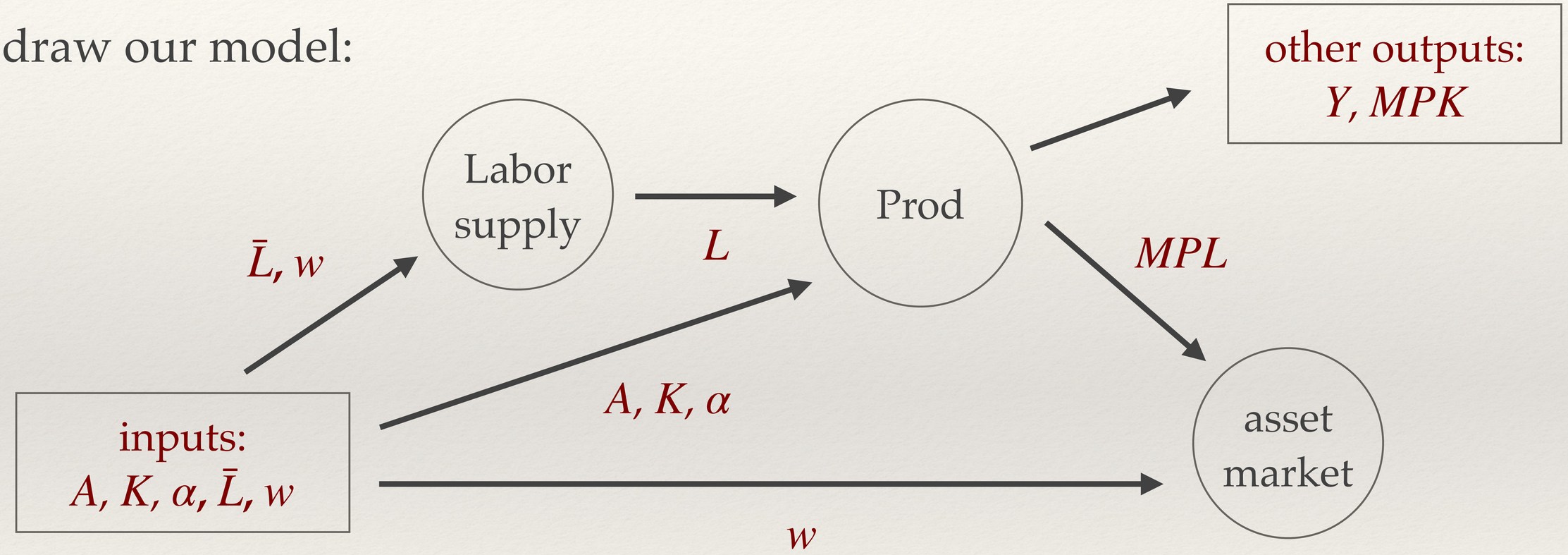
```
print(model.blocks)
print(model.inputs)
print(model.outputs)
```

✓ 0.0s

```
[<SimpleBlock 'labor_supply'>, <SimpleBlock 'prod'>, <SimpleBlock 'labor_mkt
['A', 'K', 'alpha', 'Lbar', 'w']
['Y', 'MPK', 'MPL', 'L', 'labor_mkt']
```

Directed Acyclic Graphs (DAGs)

- Can draw our model:



- The toolkit does this **automatically**, placing and evaluating blocks in the right order

Solving the model

- We can play with this model as a whole
- For instance, we can calibrate many parameters to hit targets using .solve_steady_state()
- Then, with the resulting steady state, we can use .solve_jacobian(), finding how various outcomes (Y, L, w) respond to exogenous parameters (A, $\bar{L}$), when we solve for the market-clearing w
- Under the hood, the toolkit is building a linear system using the chain rule, and solving it!

```
ss = model.solve_steady_state(calibration,
                              unknowns={'A': 1, 'alpha': 0.5, 'Lbar': 1, 'w': 1},
                              targets={'Y': 1, 'L': 1, 'MPK': 0.1, 'labor_mkt': 0})
ss
```

✓ 0.0s

```
<SteadyStateDict: ['A', 'K', 'L', 'alpha', 'Lbar', 'w', 'Y', 'MPK',
```

```
J = model.solve_jacobian(ss,
                         inputs=['A', 'Lbar'], outputs=['Y', 'L', 'w'],
                         unknowns=['w'], targets=['labor_mkt'], T=400)
J
```

✓ 0.0s

```
<JacobianDict outputs=['w', 'Y', 'L'], inputs=['A', 'Lbar']>
```

```
J['Y', 'A'][:3, :3].round(3)
```

✓ 0.0s

```
array([[2.176, 0.   , 0.   ],
       [0.   , 2.176, 0.   ],
       [0.   , 0.   , 2.176]])
```

This gets harder without a toolkit…

- The overall Jacobian of Y to A, imposing market clearing, that we saw includes:
  - The direct sensitivity of Y to A
  - The sensitivity of Y to L, combined with the sensitivity of L to w, combined with the sensitivity of w to A under market clearing
- On the right we do this manually, but it takes more effort (and we're cheating by taking the sensitivity of w to A we already calculated)!
- As models become more complicated, probability of a mistake with manual calculation $\longrightarrow 1$

```
Y_AL = prod.jacobian(ss, inputs=['L', 'A'], outputs=['Y'])
Y_A, Y_L = Y_AL['Y', 'A'], Y_AL['Y', 'L']
L_w = labor_supply.jacobian(ss, inputs=['w'], outputs=['L'])['L', 'w']
L_A = L_w @ J['w', 'A']
Y_AL = Y_L @ L_A + Y_A
Y_AL[:3, :3].round(3)
```

✓ 0.0s

```
array([[2.176, 0.   , 0.   ],
       [0.   , 2.176, 0.   ],
       [0.   , 0.   , 2.176]])
```

Concept 4: HetBlocks

- **HetBlocks** represent heterogeneous agents
- Define with backward iteration function
- Takes in the forward-looking "backward" variable (here, Va_p) plus other inputs
- Returns this period's backward variable (here, Va), plus the policy variable (here, assets 'a') and other variables of interest
- Supply key information about hetblock, including the matrix or matrices (here, Pi) giving exogenous state transitions, and an initializer function (here, hh_init)

```
def hh_init(a_grid, y, r, eis):
    coh = (1 + r) * a_grid[np.newaxis, :] + y[:, np.newaxis]
    Va = (1 + r) * (0.1 * coh) ** (-1 / eis)
    return Va

@het(exogenous='Pi', policy='a', backward='Va', backward_init=hh_init)
def hh(Va_p, a_grid, y, r, beta, eis):
    uc_nextgrid = beta * Va_p
    c_nextgrid = uc_nextgrid ** (-eis)
    coh = (1 + r) * a_grid[np.newaxis, :] + y[:, np.newaxis]
    a = interpolate.interpolate_y(c_nextgrid + a_grid, coh, a_grid)
    misc.setmin(a, a_grid[0])
    c = coh - a
    Va = (1 + r) * c ** (-1 / eis)
    return Va, a, c
```

[This is code for the built-in sj.hetblocks.hh_sim.hh model in the toolkit, which we'll use most often. It's very similar to our fast SIM code, just with slightly different function names, and it doesn't apply Pi since the toolkit takes care of that.]

HetBlocks continued

- All methods can be applied to HetBlocks
- So under the hood, the toolkit can do a **ton**: solve for steady state, iterate backward and forward nonlinearly, apply fake news algorithm to get Jacobian, etc…
- For now, we allow arbitrarily many discrete exogenous states (transitions given by 'exogenous'), then one or two endogenous continuous states (by 'policy')
  - Exogenous states ordered first, then endogenous states, in dimensions of return variables
- Extension under development ("StageBlock") that's far more general…

```
def hh_init(a_grid, y, r, eis):
    coh = (1 + r) * a_grid[np.newaxis, :] + y[:, np.newaxis]
    Va = (1 + r) * (0.1 * coh) ** (-1 / eis)
    return Va

@het(exogenous='Pi', policy='a', backward='Va', backward_init=hh_init)
def hh(Va_p, a_grid, y, r, beta, eis):
    uc_nextgrid = beta * Va_p
    c_nextgrid = uc_nextgrid ** (-eis)
    coh = (1 + r) * a_grid[np.newaxis, :] + y[:, np.newaxis]
    a = interpolate.interpolate_y(c_nextgrid + a_grid, coh, a_grid)
    misc.setmin(a, a_grid[0])
    c = coh - a
    Va = (1 + r) * c ** (-1 / eis)
    return Va, a, c
```

```
print(hh)
print(hh.inputs)
print(hh.outputs)
```

✓ 0.0s

```
<HetBlock 'hh'>
['a_grid', 'y', 'r', 'beta', 'eis', 'Pi']
['A', 'C']
```

Brief addendum: "hetinputs"

- The default "hh" HetBlock on the last slide takes a vector "y" of incomes
- In our basic GE model, these incomes are the product of after-tax aggregate income Z_t and labor endowment e
- Could write a new HetBlock with Z and e as inputs rather than y …
- … but more convenient to take the existing HetBlock, and add a new function that remaps y to other inputs
- Called a "hetinput" (also have "hetoutput")

```
def income(e, Z):
    y = Z * e
    return y

hh = hh.add_hetinputs([income])
print(hh.inputs)
```

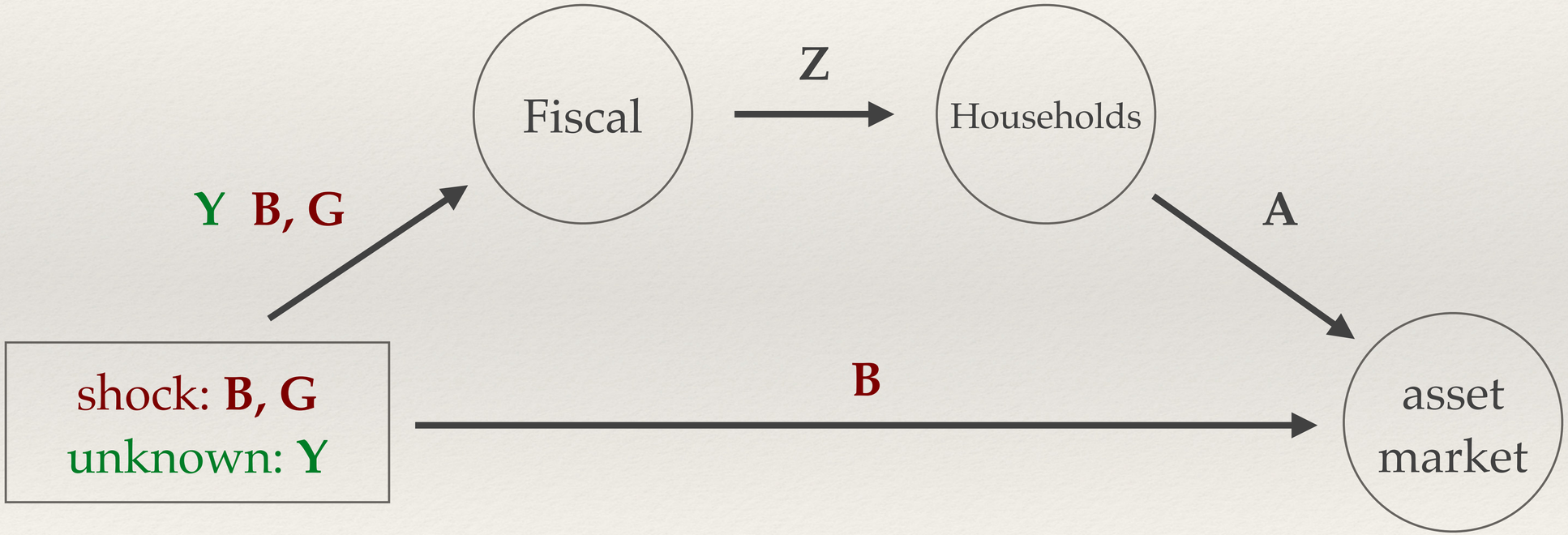
✓ 0.0s

```
['a_grid', 'r', 'beta', 'eis', 'Pi', 'e', 'Z']
```

Application 1: redo fiscal policy exercise

Solving for output response of shocks

- Holding *r* and other parameters constant, want **Y** response to **B, G** shocks



- Last lecture we derived this analytically, now let's write generic SSJ

Writing in SSJ toolkit

```
@sj.simple
def fiscal(B, r, G, Y):
    T = (1 + r) * B(-1) + G - B
    Z = Y - T
    return T, Z

@sj.simple
def mkt_clearing(A, B, Y, C, G):
    asset_mkt = A - B
    goods_mkt = Y - C - G
    return asset_mkt, goods_mkt

model = sj.combine([hh, fiscal, mkt_clearing])
```

```
from calibration import make_calibration
calib, e = make_calibration(lowA=True)
calib.update({'e': e, 'B': 4,
              'T': 0.3, 'Y': 1,
              'G': 0.3 - ss['r']*4})

ss = model.steady_state(calib)
assert np.isclose(ss['asset_mkt'], 0, atol=1E-7)
assert np.isclose(ss['goods_mkt'], 0, atol=1E-7)
```

Since we already calculated a calibration outside the toolkit, we'll use that here, then verify that markets clear when we embed that into GE (note: bonds here are 100% of annual GDP, 400% of quarterly.)

Alternatively, could have used toolkit for everything.

Now solution is a single line of code!

- In our model, we're solving for an unknown output sequence **Y** to hit asset market clearing, given a shock to the path of bonds **B**
  - which induces a change in transfers **T**
  - single line of SSJ, using solve_impulse_linear!
  - can get the nonlinear solution for same shock, just solve_impulse_nonlinear!
- Finally, verify that it's the same as our direct linear solution with the "A" matrix.

```
imp_linear = model.solve_impulse_linear(ss,
        unknowns=['Y'], targets=['asset_mkt'],
        inputs={'B': dB})
imp_linear['Y'][:5].round(3)
```

✓ 1.7s

```
array([0.043, 0.042, 0.041, 0.04 , 0.038])
```

```
imp_nonlin = model.solve_impulse_nonlinear(ss,
        unknowns=['Y'], targets=['asset_mkt'],
        inputs={'B': dB}, verbose=False)
imp_nonlin['Y'][:5].round(3)
```

✓ 3.7s

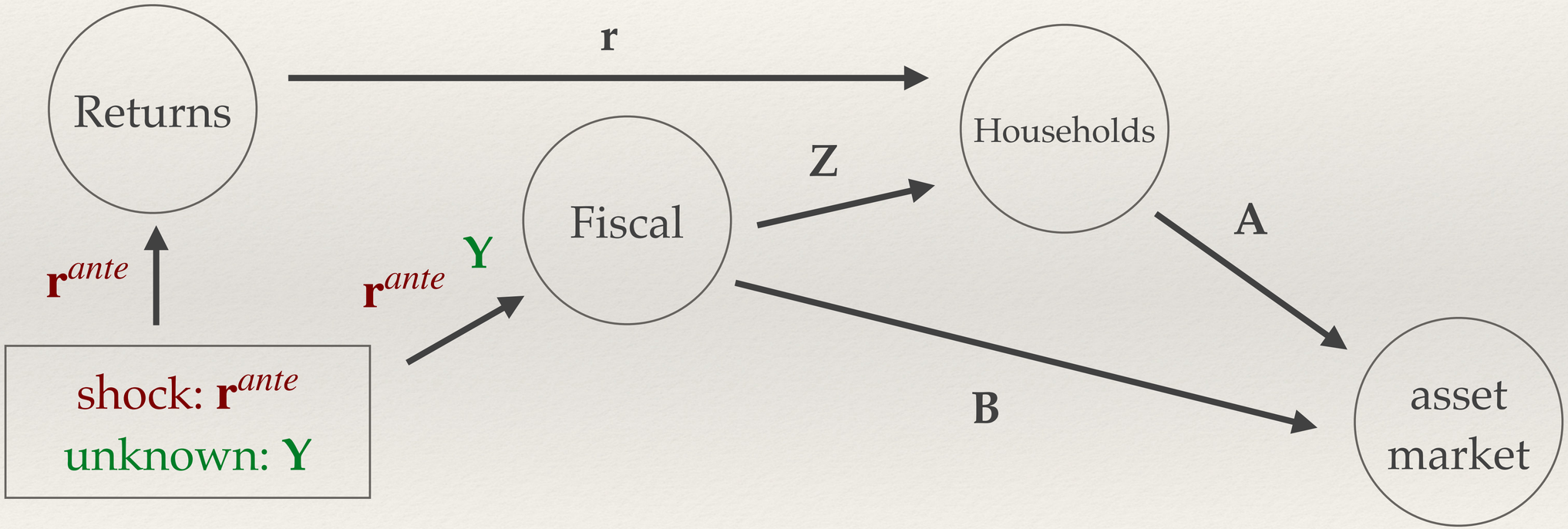
```
array([0.041, 0.04 , 0.039, 0.038, 0.037])
```

```
A = hh.jacobian(ss, inputs=['Z'],
                outputs=['A'], T=400)['A', 'Z']
dY = np.linalg.solve(A, dB) + dT
assert np.allclose(dY, imp_linear['Y'])
```

Application 2: redo monetary policy exercise

Solving for output response of shocks

- Now we want to shock ex-ante real interest rates as in the last lecture



- Two complications: we need to shift $\mathbf{r}^{ante}$ to our ex-post **r** that household uses, and **B** is also (slightly) endogenous here, based on the rule that holds $(1 + r_t)B_t$ fixed

Writing in SSJ toolkit

```
@sj.simple
def returns(r_ante):
    r = r_ante(-1)
    return r

@sj.simple
def fiscal(r_ante, G, Y):
    B = ss['B'] * (1 + ss['r']) / (1 + r_ante)
    T = (1 + r_ante(-1)) * B(-1) + G - B
    Z = Y - T
    return T, Z, B

model = sj.combine([returns, fiscal, mkt_clearing, hh])
ss = model.steady_state({**ss, 'r_ante': ss['r']})
```

The market-clearing block is unchanged from before, and we can reuse essentially the same steady state as in fiscal case.

```
# scale shock as a 1pp (annualized) cut
# note vector "dr" from lecture 4 is dlog(1+r_ante)
dr = -0.0025 * 0.9**np.arange(T) / (1 + ss['r'])
imp_linear = model.solve_impulse_linear(ss,
                    unknowns=['Y'], targets=['asset_mkt'],
                    inputs={'r_ante': (1+ss['r'])*dr})
(imp_linear['Y'][:5]).round(3)
```

✓ 2.7s

```
array([0.022, 0.02 , 0.018, 0.017, 0.015])
```

```
imp_nonlin = model.solve_impulse_nonlinear(ss,
        unknowns=['Y'], targets=['asset_mkt'],
        inputs={'r_ante': (1+ss['r'])*dr}, verbose=False)
(imp_nonlin['Y'][:5]).round(3)
```

✓ 3.2s

```
array([0.022, 0.02 , 0.018, 0.017, 0.015])
```

No visible nonlinearity for this shock!

Final application:
add Taylor rule to fiscal policy case

Write blocks for NKPC, Taylor rule

```
@sj.simple
def fiscal(B, r, G, Y):
    # note: taking B as exogenous again
    T = (1 + r) * B(-1) + G - B
    Z = Y - T
    return T, Z

beta_mean = ss['beta'].mean()
@sj.solved(unknowns={'pi': (-0.5, 0.5)},
           targets=['nkpc_resid'])
def pricing(pi, kappa, Y):
    nkpc_resid = (pi - kappa*(Y - ss['Y'])
                      - beta_mean*pi(+1))
    return nkpc_resid

@sj.simple
def taylor(pi, phi_pi, r_ante):
    i = ss['r'] + phi_pi * pi
    r_resid = i - pi(+1) - r_ante
    return r_resid
```

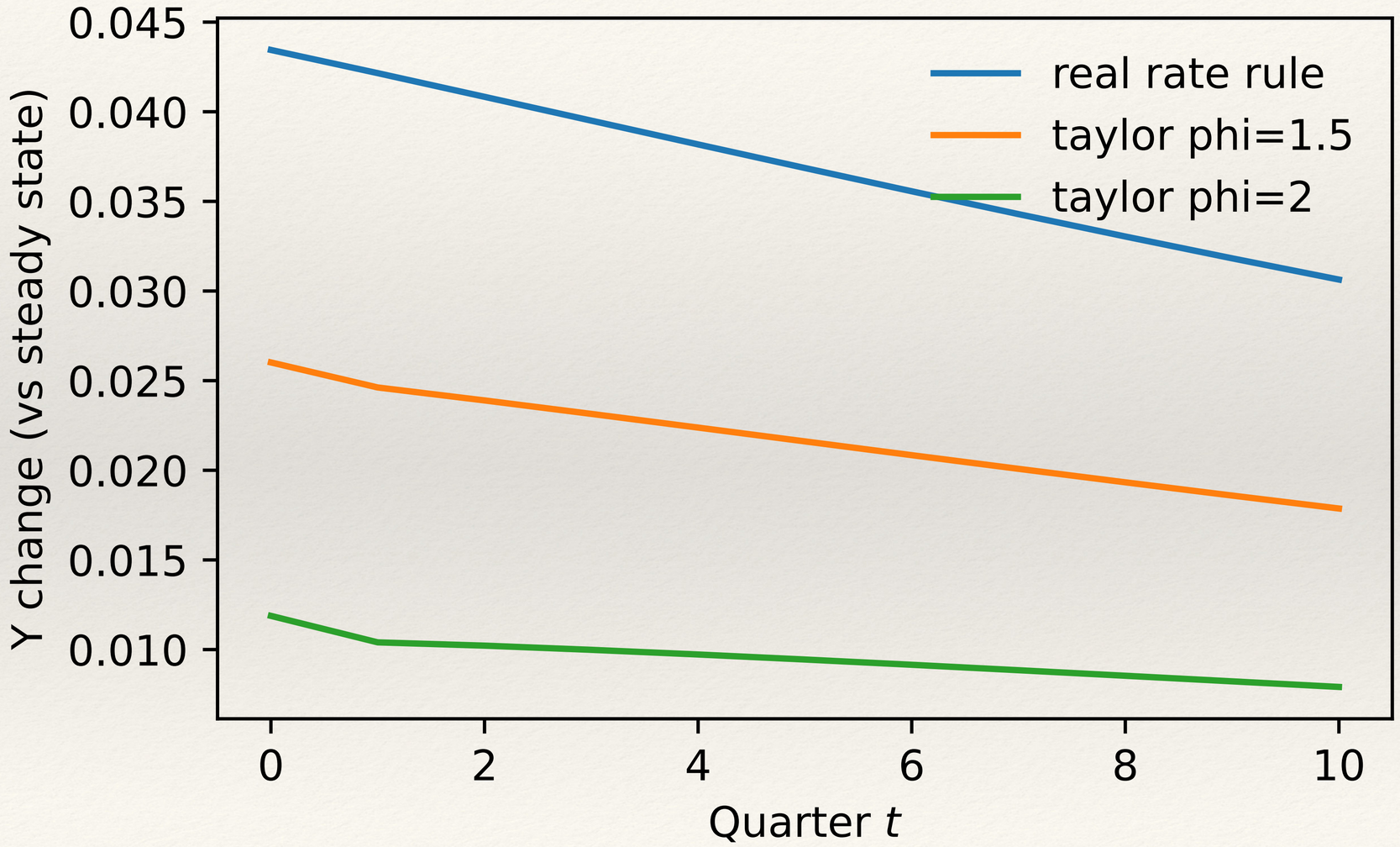
```
model = sj.combine([hh, returns, fiscal,
                    pricing, taylor, mkt_clearing])

ss = model.steady_state({**ss,
        'kappa': 0.01, 'phi_pi': 1.2})

dY_phi_12 = model.solve_impulse_linear(ss,
            unknowns=['Y', 'r_ante'],
            targets=['asset_mkt', 'r_resid'],
            inputs={'B': dB})['Y']
```

Now need to define a “pricing” block, which solves for pi to hit the NKPC, and a Taylor rule. In the main loop, solve for another unknown, r_ante, to be consistent with the Taylor rule and Fisher equation.

Effect of tax cut shock by monetary policy



Conclusion

Concluding thoughts

- We can always work manually with Jacobians, but the math quickly gets complicated!
  - Lots of variables to handle, manually doing the fake news algorithm, manually doing the chain rule…
- The SSJ toolkit offers a systematic way to avoid this drudgery
- It's also a useful **language** in which to think about writing models
- Still some bugs and quirks that we'll work to fix, but with AI it's easier than ever to use